



MUHAMMAD ILHAM FIRMANSYAH

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github.com/Ilhamcom23 | <https://ilhamcom23.github.io/portfolio/>

Jl. Raya Tulangan No. 27, East Tulangan, Tulangan, Tulangan District, Sidoarjo Regency, East Java 61273
Informatics Engineering student at Jember State Polytechnic with technical competencies in web development, IoT, and artificial intelligence implementation. He has a solid portfolio track record, including the development of a Geographic Information System (GIS) for hydroponic farming mapping in Pasuruan, Computer Vision integration (PaddleOCR) for automatic label detection, and a Smart Greenhouse innovation tested through the Student Creativity Program (PKM). He is proficient in Python and Laravel with a focus on creating efficient and impactful technology solutions.

Jenjang Pendidikan

SMKN 8 JEMBER - Jl. Pelita No. 27, Sidomekar Village, Semboro District, Jember Regency, East Java, Indonesia. Jul 2018 - Aug 2021

Senior High School, Software engineering,, 3.80/4.00

- Sertifikat Magang Kantor Kominfo Jember
- Sertifikat OF Competency MASPION IT 2021
- Sertifikat OF Competency APTIKNAS 2021
- Sertifikat OF Competency AMIGROUP 2021

Politeknik Negeri Jember - Jl. Mastrip PO.BOX 164, East Krajan, Tegalgede, Sumpstersari District, Jember Regency, East Java, 68124 Aug 2022 - Jun 2026

Bachelor of Informatics Engineering, 3.66/4.00

- Participant Literasi Digital 2023
- Participant Coding Camp Exploring Front-End Developer : Wonders and Opportunities 2023
- Sertifikat : Dicoding Camp Web Component menggunakan Lit 2023
- Participant Kompetisi KMIPN VI 2024 Kategori Cipta Inovasi
- Participant Kompetisi GEMASTIK 2024 Kategori Perangkat Lunak

Pengalaman Kerja

Jember Regency Communication and Informatics Office (Diskominfo) - Jember, East Java Jan 2020 - Jun 2020

Full Stack Web Developer

Government agency responsible for ICT infrastructure, public information systems, and digital services in Jember Regency

- Managed and updated the official government website to ensure accurate, up-to-date, and user-friendly content.
- Implemented interface improvements and responsive design, increasing website traffic by approximately 30% within six months.
- Organized and managed digital data and documentation of institutional activities, improving data retrieval efficiency by 40%.
- Assisted in resolving technical issues related to networking and software systems to support daily operations.

PT Masrur And Son - Sidoarjo, East Java Mar 2024 - Jun 2024

Full Stack Web Developer

Private manufacturing company focusing on warehouse operations and inventory management systems

- Developed a web-based Smart Warehouse (SmartGudang) application to automate inventory management processes.
- Implemented real-time stock recording and monitoring features, improving inventory management efficiency by 30%.
- Increased inventory data accuracy to 95%, reducing manual errors and supporting faster decision-making.
- Collaborated with operational teams to ensure system usability and alignment with warehouse workflows.

PT Telkom Indonesia (Tbk) - Jember, East Java Aug 2025 - Dec 2025

Intership

Indonesia's leading state-owned telecommunications company providing digital infrastructure and services nationwide

- Developed an AI-based system to detect Optical Distribution Points (ODP) using object detection techniques.
- Integrated OCR technologies (Tesseract / PaddleOCR) to automatically extract text from ODP labels.
- Built an end-to-end web pipeline enabling users to upload images and receive real-time detection results and extracted label information.
- Supported digital transformation initiatives by applying computer vision and automation technologies to field operations.

Pengalaman organisasi

general Secretary

A youth organization focused on social, religious, and youth empowerment initiatives in Jember, emphasizing character development through educational programs and community service.

- Systematically manage incoming and outgoing mail, ensuring timely distribution, and archiving important documents to maintain order and easy access to information.
- Serve as a liaison between Mosque Youth members and external parties, facilitating effective communication.
- Coordinate meeting and event schedules to ensure organizational activities run smoothly and according to plan.
- Manage member databases, including regularly updating contact information to ensure data accuracy.
- Support the implementation of organizational programs with high administrative efficiency.

Keterampilan, Prestasi & pengalaman lain

- **Technical Skills:** Python, PHP, HTML5, CSS3, SQL, C++ (Arduino/IoT), Laravel, Flask, REST API, Computer Vision (YOLO, CNN), OCR (PaddleOCR, Tesseract), IoT Systems, Image Processing.
- **Projects** (2024): Smart Greenhouse IoT System – Developed an IoT-based greenhouse system integrating temperature, humidity, soil pH, water level, and light intensity sensors with real-time data analysis.
- **Projects** (2024): Smart Warehouse System – Web & Mobile – Built a warehouse management system to accelerate inventory recording and monitoring through web and mobile applications
- **Projects** (2025): School / Company Profile Website – Designed a responsive role-based company profile website with admin content management and optimized UI/UX.
- **Languages** 🌐: Bahasa Indonesia (Native), English (Intermediate).
- **Achievements** 🏆 (2022): Halal Certificate Companion – MUI Sidoarjo